Mako-Style Speculative Execution vs. Replication-Coupled Transactions

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INTRODUCTION

Why replication placement matters

Distributed transactions coordinate work across shards and replicate outcomes for durability. Replication-coupled designs keep execution workers occupied while replication completes.

Mako-style speculation lets certified transactions release execution capacity while durability work continues on separate streams.

Research question

In a controlled transaction-path model, how does increasing network RTT affect Mako-style speculative execution compared with replication-coupled execution?

Evidence boundary

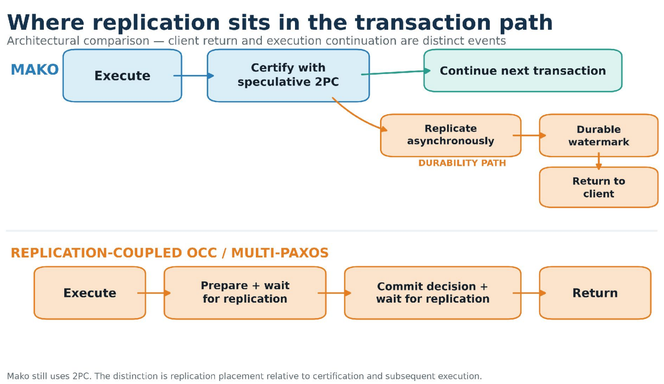
Implementation validation and original simulation results are reported separately. Simulation values are not measurements of the Mako implementation.

APPROACH

What we compared

- Mako-style: release the execution worker after certification; replicate on a separate durability stream.
- Coupled path: hold the worker through prepare and commit replication rounds.

In the model, both clients still wait for durability before completion.



Mako still uses 2PC. The distinction is replication placement relative to certification and subsequent execution.

IMPLEMENTATION VALIDATION

Official implementation checks

- Native ELF64 x86-64 binaries built
- Five semantic checks passed
- Replicated-Mako and replication-coupled smoke tests passed
- Replication activity across voters verified

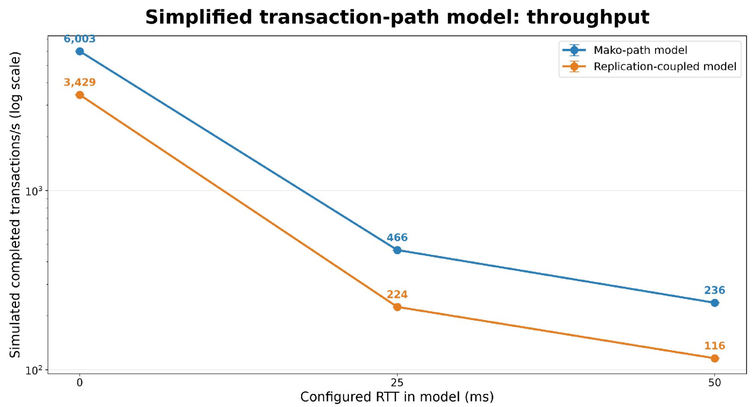
Smoke tests establish operability and protocol exercise—not performance superiority.

Seeded simulation design

- 18 trials: 2 paths × 3 configured RTTs × 3 deterministic seeds
- RTT: 0, 25, and 50 ms; 96 clients; 12 execution workers
- 5 s discarded warm-up; 20 s measurement window
- Identical local execution and certification distributions
- 12 separate durability streams for the Mako-style path

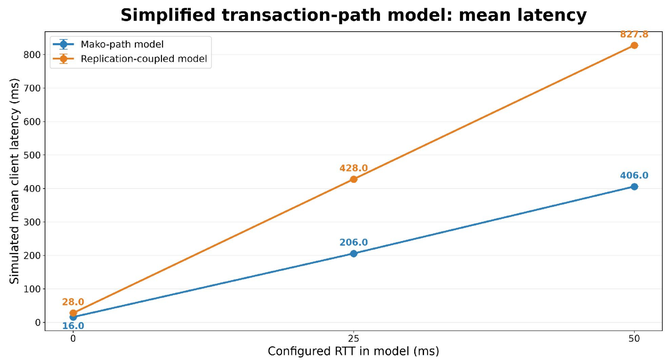
RESULTS

IN OUR MODEL: 75.1%–107.8% more simulated completions/s across 0–50 ms RTT



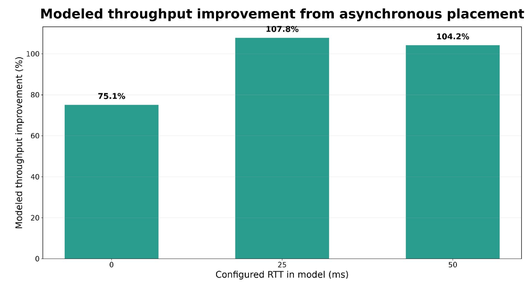
SIMULATED STRUCTURAL MODEL — not measurements of Mako, the baseline implementation, or the OSDI testbed

Discrete-event simulation — not implementation benchmark results.



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SIMULATED STRUCTURAL MODEL — improvement follows from the stated model assumptions; it is not an implementation result

Discrete-event simulation — not implementation benchmark results.

CONCLUSION

- In our model, separate durability streams let workers overlap later transactions.
- RTT harms both paths; worker retention exposes the coupled path more.
- Modeled result—not real Mako performance.

Limitations

Simulation results depend on a synthetic workload and configured service times; contention and failures are omitted. The native matrix was blocked by baseline runtime failures.

FUTURE WORK

- Repair baseline TPC-C population/runtime defects and rerun with native binaries.
- Calibrate on native multi-machine x86-64 Linux.
- Add contention, failures, recovery, batching, and additional workloads.

References: Shen et al., OSDI '25 • makodb/mako @ f5567003… • Original seeded simulator, configuration, CSVs, and retained validation logs